

Fourier Series		Symmetries	
Conjugate	$\overline{x(t)} \rightarrow \overline{a_{-k}}$	$x(t)$ Real $\iff a_k$ Hermitian	$x(t) = \overline{x(-t)} \rightarrow a_k = \overline{a_{-k}}$ $x(t) = \overline{x(t)} \rightarrow a_k = \overline{a_{-k}}$
Time inversion	$x(-t) \rightarrow a_{-k}$	$x(t)$ Imag $\iff a_k$ anti-Herm	$x(t) = -\overline{x(-t)} \rightarrow a_k = -\overline{a_{-k}}$ $x(t) = -\overline{x(t)} \rightarrow a_k = -\overline{a_{-k}}$
Time scale	$x(\alpha t) \rightarrow a_k \alpha > 0$ $T' = T/\alpha, w' = \alpha w$	$x(t)$ Even $\iff a_k$ Even Real/imag preserved	$x(t) = x(-t) \rightarrow a_k = a_{-k}$
Traslation	$x(t + \beta) \rightarrow e^{jk w_0 \beta} a_k$	$x(t)$ Odd $\iff a_k$ Odd Real/imag switched	$x(t) = -x(-t) \rightarrow a_k = -a_{-k}$
Modulation	$e^{jM w_0 t} \rightarrow a_{k-M}$		
Derivation	$\frac{dx(t)}{dt} \rightarrow jk w_0 a_k$		

Fourier Transform			
Rect Sinc	$rect(t) \rightarrow sinc(w/2\pi)$	$rect(t) \rightarrow sinc(w/2\pi)$	
Exponentials	$e^{-\sigma t} u(t) \rightarrow \frac{1}{\sigma + jw}$	$e^{\sigma t} u(-t) \rightarrow \frac{1}{\sigma - jw}$	$e^{-\sigma t } \rightarrow \frac{2\sigma}{\sigma^2 + w^2}$
Fractions	$1/t \rightarrow -j\pi sign(w)$	$\frac{1}{1+t^2} \rightarrow \pi e^{- \omega }$	
1 dirac unit	$1 \rightarrow 2\pi\delta(w)$	$\delta(t) \rightarrow 1$	$u(t) \rightarrow \frac{1}{jw} + \pi\delta(w)$
Fourier Series	$e^{jw_0 t} \rightarrow 2\pi\delta(w - w_0)$	$\sum a_k e^{jw_0 t} \rightarrow 2\pi \sum a_k \delta(w - k w_0)$	
Sin Cos	$sin(w_0 t) \rightarrow j\pi(\delta(w + w_0) - \delta(w - w_0))$	$cos(w_0 t) \rightarrow \pi(\delta(w + w_0) + \delta(w - w_0))$	
Time Shift	$x(t + a) \rightarrow e^{jwa} X(w)$	$e^{jw_0 t} x(t) \rightarrow X(w - w_0)$	
Modulation	$x(at) \rightarrow \frac{1}{ a } X(w/a)$		
Product Convolution	$x * y(t) \rightarrow X(w) \times Y(w)$	$x(t) \times y(t) \rightarrow \frac{1}{2\pi} X(w) * Y(w)$	
Derivate	$x'(t) \rightarrow jw X(w)$	$x^{(n)}(t) \rightarrow (jw)^n X(w)$	
Derivative in t	$tx(t) \rightarrow j \frac{d}{dw} X(w)$	$t^n x(t) \rightarrow j \frac{d^n}{dw^n} X(w)$	
Integral	$\int_{-\infty}^t x(t) dt \rightarrow \frac{X(w)}{jw} + \pi X(0)\delta(w)$		
Duality	$X(w) _{w=t} \rightarrow 2\pi x(-w)$		

Laplace		DTFT	
$\delta(t) \rightarrow 1$		$a^n u(n) \rightarrow \frac{e^{jw}}{e^{jw} - a}$	$ a < 1$
$u(t) \rightarrow \frac{1}{s}$		$-a^n u(-n-1) \rightarrow \frac{1}{1 - a e^{-jw}}$	$ a > 1$
$tu(t) \rightarrow \frac{2}{s^3}$	$t^n u(t) \rightarrow \frac{n!}{s^{k+1}}$	$a^{ n } \rightarrow \frac{1-a^2}{1+a^2-2acos(w)}$	$ a < 1$
$e^{\sigma t} u(t) \rightarrow \frac{1}{s-\sigma}$	$te^{\sigma t} u(t) \rightarrow \frac{1}{(s-\sigma)^2}$	discrete rect $\rightarrow \frac{sin(N+\frac{1}{2})w}{sin(w/2)}$	
$cos(wt)u(t) \rightarrow \frac{s}{s^2+w^2}$	$e^{\sigma t} cos(wt)u(t) \rightarrow \frac{s-\sigma}{s^2+w^2}$		
$sin(wt)u(t) \rightarrow \frac{w}{s^2+w^2}$	$e^{\sigma t} sin(wt)u(t) \rightarrow \frac{w}{(s-\sigma)^2+w^2}$		

TDF Symmetries	DTFT Symmetries
$x(t)$ real $\iff X(w)$ real	$x(t)$ real $\iff X(e^{jw})$ Hermitian

TDF Symmetries	DTFT Symmetries
$x(t)$ even $\iff X(w)$ even	$x(t)$ even $\iff X(e^{jw})$ even
$x(t)$ odd $\iff X(w)$ odd	$x(t)$ odd $\iff X(e^{jw})$ odd
$x(t)$ real even $\iff X(w)$ real even	$x(t)$ real even $\iff X(e^{jw})$ real even
$x(t)$ real odd $\iff X(w)$ real odd	$x(t)$ real odd $\iff X(e^{jw})$ real odd

ideal low pass	ideal high pass	ideal passa banda
$H(w) = \text{rect}(\frac{w}{2w_c})$	$H(w) = 1 - \text{rect}(\frac{w}{2w_L})$	$H(w) = \text{rect}(\frac{w}{2w_H}) + \text{rect}(\frac{w}{2w_L})$

- Laplace Derivative

$$\mathcal{L}\left(\frac{d}{dt}x(t)\right) = sX(s) - x(0-)$$

$$\mathcal{L}\left(\frac{d^n}{dt^n}x(t)\right) = s^n X(s) - s^{n-1}x(0-) - s^{n-2}x'(0-) - \dots - x^{(n-1)}(0-)$$

- By Shannon it is possible to reconstruct $x(t)$ with the following formula $x(t) = \sum_{k=-\infty}^{\infty} x(kT) \text{sinc}(\frac{t-kT}{T})$

if $w_s > 2w_M$

- Rect Triangle Convolution: $\text{tri}(\frac{t}{T}) = \frac{1}{T} \text{rect}(\frac{t}{T}) * \text{rect}(\frac{t}{T})$

- By same logic: $\mathcal{F}(\text{sinc}(\frac{t}{T})^2) = T \text{tri}(\frac{wT}{2\pi})$

- $T = \frac{2\pi}{w}, f = \frac{1}{T} = \frac{w}{2\pi}$

- $\sin(x) = -\sin(-x), \cos(x) = \cos(-x)$

- If an LTI system is discrete time $\implies H(w)$ is periodic

- LTI is causal if $h(t) = 0 \forall t < 0$

- DTFT if $X(w) = \text{rep}_{2\pi}(f(w))$ then I can use ant DTFT to find $x(t)$

- Convolutional representation of a periodic signal $\text{rep}_T(x(t)) = \sum_{k=-\infty}^{\infty} x(t - kT) = \sum_{k=-\infty}^{\infty} \delta(t - kT)x(t)$

- Delta and Conv

$$x(t + \alpha) * y(t + \beta) = x(t + \alpha + \beta) * y(t)x(t) * y(t + \alpha + \beta)$$

$$f(t - \beta)\delta(t - \alpha) = f(\alpha - \beta)\delta(t - \alpha)$$

$$f(t - \beta) * \delta(t - \alpha) = \int f(\alpha - \beta)\delta(t - \alpha)d\tau = f(t - \alpha - \beta)$$

- SDF

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jw_k t}$$

$b_k = H(w_k)a_k$, that means it can only contain frequencies already present in $x(t)$

??? $x(t) = \text{rep}_T(\text{rect}(t/t_0)), a_k = \frac{t_0}{T} \text{sinc}(\frac{w_k t_0}{2\pi})$

??? $\text{trap} = \text{rect}(t) * \text{rect}(\frac{t}{T})$ is trap of length T?